Fiscal Year:	FY 2014	**Task Last Updated:**	FY 12/30/2014
PI Name:	Voorhees, Peter W. Ph.D.		
Project Title:	Coarsening of Dendritic Solid-Liquid Mixtures: The Low Volume Fraction Limit		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
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PI Email:	p-voorhees@northwestern.edu	**Fax:**	FY 847-491-7820
PI Organization Type:	UNIVERSITY	**Phone:**	847-491-7815
Organization Name:	Northwestern University		
PI Address 1:	Department for Materials Science and Engineering		
PI Address 2:	2220 Campus Dr		
PI Web Page:			
City:	Evanston	**State:**	IL
Zip Code:	60208-0893	**Congressional District:**	9
Comments:			
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Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
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Flight Assignment:	ISS NOTE: End date is now 3/4/2014 per NSSC information (Ed., 3/10/14) NOTE: New end date of 9/27/2014, per PI and NSSC information (Ed., 9/3/13)		
Key Personnel Changes/Previous PI:			
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Task Description:	Dendrites form during solidification into an undercooled melt. These dendrites possess secondary and sometimes even ternary arms. While the tip radius and tip velocity of the dendrite are set by the growth conditions, the sidebranches behind the tip undergo a coarsening process under nearly isothermal conditions. This coarsening process sets the final arm thicknesses and distances between dendrite arms in the solidified structure - almost independent of the length-scale given by the dendrite tip. Unfortunately, the morphological evolution of these secondary dendrite arms remains poorly understood. Since there is a close relationship between the size scale of the coarsened dendritic structure and the mechanical properties of the material, a greater understanding of this process will have important practical implications. The reduced gravitational accelerations found during processing on the International Space Station can allow for unprecedented insights into coarsening processes of solid-liquid mixtures in which the volume fraction of solid dendrites is small. In a microgravity environment it will be possible to follow the evolution of the morphology of the mixtures in the diffusive limit where the concentration field is set by the curvatures of the solid-liquid interfaces. It will also be possible to determine the evolution of the topology of the solid-liquid mixtures. For example, when secondary dendrite arms detach from the primary dendrite stem they will not sediment, and it will thus be possible to determine the rate at which this detachment process occurs and its spatial location. The topology and morphology of these dendritic mixtures will be determined using three-dimensional reconstructions and insights into the mechanisms of coarsening will be provided by phase field simulations that employ these reconstructions as initial conditions. These studies will provide insights into the dynamics of morphological evolution of these systems and will point the way to the formulation of models of this technologically important coarsening process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Dendritic solid-liquid mixtures are ideal topologically complex systems with which to study coarsening. The microstructure is on a size scale that is amenable to experimental analysis using three-dimensional reconstructions and the thermophysical properties of many alloy systems are reasonably well known. Moreover, the coarsening of dendritic solid-liquid mixtures has major technological implications since it occurs in virtually all casting processes. In this case, a dendritic structure forms quickly and a so-called "mushy zone" is created. This mushy zone can exist often for long periods of time during which virtually every dendrite undergoes coarsening. The length-scales of the dendrites in the mushy zone, and thus those present after complete solidification, are determined by the coarsening process. These length scales set the microsegregation pattern and are intimately related to the mechanical properties of the ingot. Understanding the dynamics of coarsening will enable a close connection to be made between solidification conditions and the properties of the material.
Task Progress:	As the focus is on low volume fraction limit, a series of Pb-Sn alloys with volume fractions ranging from 10% to 40% were prepared. During increment 34-45 aboard the ISS, the samples were heated at 185C, 2C above the eutectic in the solid-liquid region of the phase diagram for various times (10m, 1.6h, 5.5h, 13.5h, 27h, and 48h). All samples were properly quenched except for the 5.5h sample that was slow cooled. We received the samples back on March 29, 2013. Using the samples returned to us after their flight to the ISS during increments 34-35, many analyses were performed to give us insight into the dynamics of the coarsening process. Multiple 3D reconstructions have been made to study of the morphological evolution of the solid-liquid interfaces. These were first done with the 30% samples, but work has also been started on the 20% samples. The reconstructions for the 30% samples were coarsened for 10 min., 1.6 hours, 5.5 hours, and 48 hours. These reconstructions encompass large volumes due to the methodology employed. For example, the volume of the structures are typically 4500x3600x1200µm. The size of these datasets is approximately 100 GB at full resolution. Using these large datasets, we are characterizing the mean curvature of the solid-liquid interfaces and the interfacial topology. For example, there is a significant probability of having patches of interface with cylindrical shapes, due to the stems of the dendrites, and solid spherical patches that is a result of the tips of the dendrites. Interestingly, there is also significant probability of regions of interface that are nearly flat. Using these techniques we will quantify the evolution of the microstructure during coarsening. See continuation grant "Coarsening of Dendritic Solid-Liquid Mixtures: The Low Volume Fraction Limit (NNX14AB73G)" for future reporting.
Bibliography Type:	Description: (Last Updated: 01/17/2024)